

Continuity and the Intermediate Value Theorem



The three-part continuity test

- 1 State the three conditions that must all hold for f to be continuous at $x = a$.
 - 2 Determine whether $f(x) = \frac{x^2-4}{x-2}$ (for $x \neq 2$, with $f(2) = 4$ defined separately) is continuous at $x = 2$.
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Finding values that ensure continuity

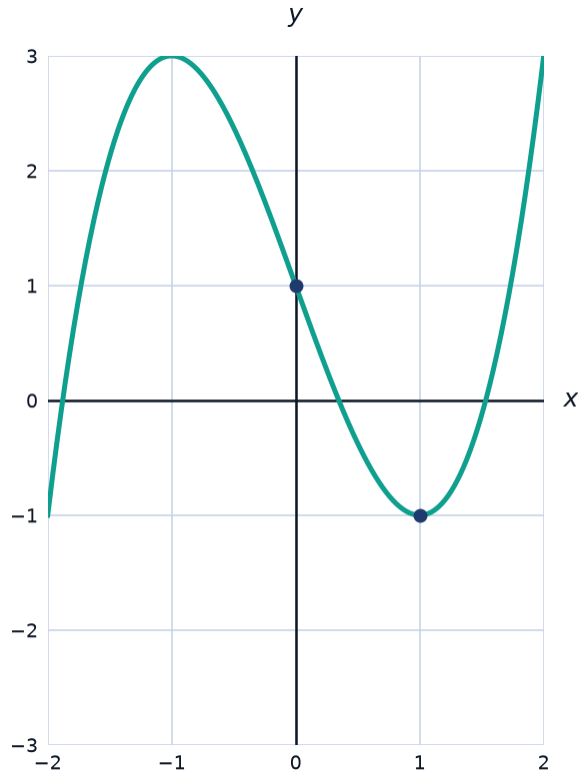
- 3 Find k so that $f(x) = kx^2$ for $x \leq 2$, and $f(x) = 5x - 2$ for $x > 2$, is continuous at $x = 2$.
 - 4 Find a and b so that $f(x) = 2$ for $x \leq -1$, $f(x) = ax + b$ for $-1 < x < 2$, and $f(x) = 6$ for $x \geq 2$, is continuous everywhere.
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Types of discontinuity

- 5 Classify the discontinuity of each function at the indicated point:
 - a $f(x) = \frac{x+3}{x^2-9}$ at $x = -3$
 - b $f(x) = \lfloor x \rfloor$ at $x = 2$
 - c $f(x) = \frac{1}{(x-1)^2}$ at $x = 1$
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The Intermediate Value Theorem

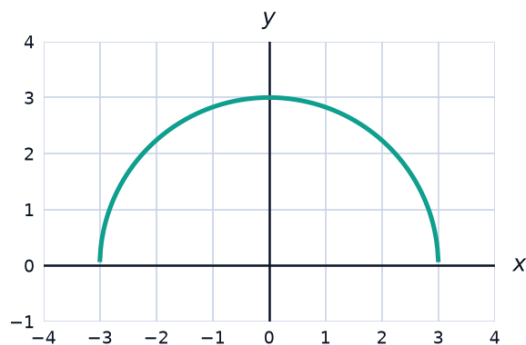
- 6 Show that $f(x) = x^3 - 3x + 1$ has a root in $(0, 1)$, using the IVT.



- 7 A continuous function g satisfies $g(1) = -3$ and $g(5) = 7$. Explain why g must take the value 0 somewhere in $[1, 5]$, and state the theorem used.
- 8 A cup of coffee's temperature (in $^{\circ}\text{C}$) is a continuous function of time, starting at 90°C and cooling to 25°C over 30 minutes. Explain, using the IVT, why the coffee must have been exactly 50°C at some point.

Continuity on an interval

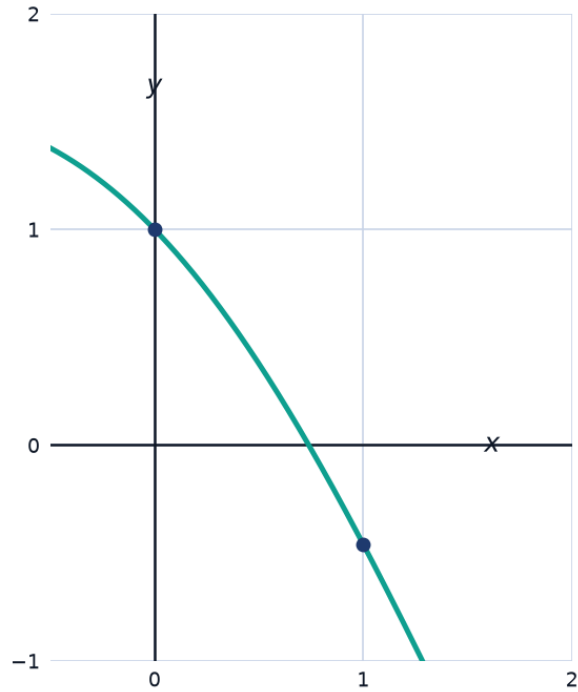
- 9 Determine the largest interval(s) on which $f(x) = \sqrt{9 - x^2}$ is continuous.



- 10 Determine all values of x where $f(x) = \frac{x+1}{x^2-x-6}$ is discontinuous, and classify each discontinuity.

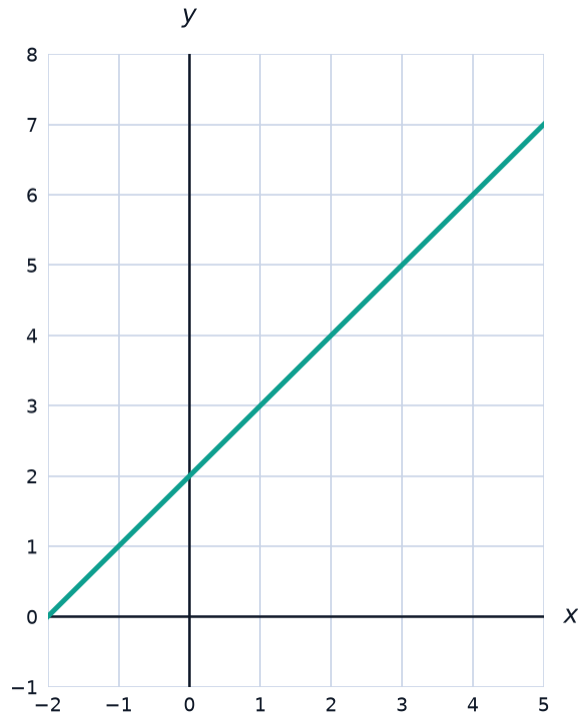
Using the IVT to approximate roots

- 11** Show that $f(x) = \cos x - x$ has a root in $(0, 1)$, then use the IVT with the midpoint to narrow the interval to a length-0.5 subinterval.



Visualizing a removable discontinuity

- 12 The graph shows $f(x) = \frac{x^2 - 4}{x - 2}$ for $x \neq 2$, simplifying to $f(x) = x + 2$. Use the graph to describe what would need to be true of $f(2)$ for the function to be continuous there.



Continuity of piecewise and combined functions

- 13 Determine whether $f(x) = |x - 3|$ is continuous at $x = 3$, checking all three conditions.
- 14 If f and g are both continuous at $x = a$, explain why $f(x) + g(x)$ and $f(x) \cdot g(x)$ must also be continuous at $x = a$, citing the relevant limit laws.

An applied IVT problem

- 15 A hiker's elevation (in meters) is a continuous function of distance walked, starting at 200 m and ending at 850 m after 5 km. Explain, using the IVT, why the hiker must have been at exactly 500 m at some point, and state why this reasoning would fail if elevation were not continuous.

Continuity with absolute value and radical functions

- 16 Determine the domain (and hence largest interval of continuity) of $f(x) = \sqrt{x^2 - 4}$.
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Multiple choice — easy 15

- 17 Find k so that $f(x) = kx$ for $x \leq 2$, and $f(x) = -x - 4$ for $x > 2$, is continuous at $x = 2$.
- a. 3
 - b. -3
 - c. -2
 - d. -4
- 18 Find k so that $f(x) = kx$ for $x \leq 2$, and $f(x) = 4x + 2$ for $x > 2$, is continuous at $x = 2$.
- a. -5
 - b. 5
 - c. 6
 - d. 4
- 19 Find k so that $f(x) = kx$ for $x \leq -2$, and $f(x) = -x + 4$ for $x > -2$, is continuous at $x = -2$.
- a. -4
 - b. -2
 - c. 3
 - d. -3
- 20 Find k so that $f(x) = kx$ for $x \leq -3$, and $f(x) = 4x + 6$ for $x > -3$, is continuous at $x = -3$.
- a. 3
 - b. 1
 - c. 2
 - d. -2

- 21 Find k so that $f(x) = kx$ for $x \leq 1$, and $f(x) = 3x - 3$ for $x > 1$, is continuous at $x = 1$.
- a. 1
 - b. 2
 - c. 0
 - d. -1
- 22 Find k so that $f(x) = kx$ for $x \leq 4$, and $f(x) = -3x + 12$ for $x > 4$, is continuous at $x = 4$.
- a. 1
 - b. 2
 - c. 0
 - d. -1
- 23 Find k so that $f(x) = kx$ for $x \leq -3$, and $f(x) = -x - 3$ for $x > -3$, is continuous at $x = -3$.
- a. -1
 - b. 1
 - c. 2
 - d. 0
- 24 Find k so that $f(x) = kx$ for $x \leq 1$, and $f(x) = 0 - 1$ for $x > 1$, is continuous at $x = 1$.
- a. 1
 - b. -1
 - c. -2
 - d. 0

- 25 Find k so that $f(x) = kx$ for $x \leq -4$, and $f(x) = -x + 12$ for $x > -4$, is continuous at $x = -4$.
- a. -3
 - b. -5
 - c. 4
 - d. -4
- 26 Find k so that $f(x) = kx$ for $x \leq -1$, and $f(x) = 0 - 1$ for $x > -1$, is continuous at $x = -1$.
- a. 0
 - b. 2
 - c. -1
 - d. 1
- 27 Find k so that $f(x) = kx$ for $x \leq 3$, and $f(x) = 2x + 3$ for $x > 3$, is continuous at $x = 3$.
- a. 4
 - b. 3
 - c. 2
 - d. -3
- 28 Find k so that $f(x) = kx$ for $x \leq 4$, and $f(x) = -4x + 8$ for $x > 4$, is continuous at $x = 4$.
- a. -1
 - b. -3
 - c. -2
 - d. 2

- 29 Find k so that $f(x) = kx$ for $x \leq -1$, and $f(x) = -2x - 3$ for $x > -1$, is continuous at $x = -1$.
- a. -1
 - b. 0
 - c. 1
 - d. 2
- 30 Find k so that $f(x) = kx$ for $x \leq -4$, and $f(x) = -3x - 8$ for $x > -4$, is continuous at $x = -4$.
- a. 1
 - b. -1
 - c. 0
 - d. -2
- 31 Find k so that $f(x) = kx$ for $x \leq -4$, and $f(x) = 2x - 12$ for $x > -4$, is continuous at $x = -4$.
- a. 4
 - b. -5
 - c. 6
 - d. 5

Multiple choice — medium 15

- 32 Find k so that $f(x) = kx^2$ for $x \leq -1$, and $f(x) = 4x - 8$ for $x > -1$, is continuous at $x = -1$.
- a. 12
 - b. -11
 - c. -12
 - d. -13

- 33 Find k so that $f(x) = kx^2$ for $x \leq 4$, and $f(x) = -2x + 7$ for $x > 4$, is continuous at $x = 4$.
- a. $-\frac{1}{16}$
 - b. $-\frac{17}{16}$
 - c. $\frac{15}{16}$
 - d. $\frac{1}{16}$
- 34 Find k so that $f(x) = kx^2$ for $x \leq 4$, and $f(x) = 4x - 1$ for $x > 4$, is continuous at $x = 4$.
- a. $-\frac{15}{16}$
 - b. $-\frac{1}{16}$
 - c. $\frac{31}{16}$
 - d. $\frac{15}{16}$
- 35 Find k so that $f(x) = kx^2$ for $x \leq -2$, and $f(x) = 5x - 9$ for $x > -2$, is continuous at $x = -2$.
- a. $\frac{19}{4}$
 - b. $-\frac{23}{4}$
 - c. $-\frac{15}{4}$
 - d. $-\frac{19}{4}$
- 36 Find k so that $f(x) = kx^2$ for $x \leq -3$, and $f(x) = -x + 1$ for $x > -3$, is continuous at $x = -3$.
- a. $-\frac{4}{9}$
 - b. $-\frac{5}{9}$
 - c. $\frac{13}{9}$
 - d. $\frac{4}{9}$

- 37 Find k so that $f(x) = kx^2$ for $x \leq 4$, and $f(x) = -3x + 9$ for $x > 4$, is continuous at $x = 4$.
- a. $-\frac{3}{16}$
 - b. $\frac{3}{16}$
 - c. $\frac{13}{16}$
 - d. $-\frac{19}{16}$
- 38 Find k so that $f(x) = kx^2$ for $x \leq 1$, and $f(x) = -5x - 1$ for $x > 1$, is continuous at $x = 1$.
- a. -6
 - b. -5
 - c. 6
 - d. -7
- 39 Find k so that $f(x) = kx^2$ for $x \leq -4$, and $f(x) = -4x + 3$ for $x > -4$, is continuous at $x = -4$.
- a. $\frac{35}{16}$
 - b. $-\frac{19}{16}$
 - c. $\frac{19}{16}$
 - d. $\frac{3}{16}$
- 40 Find k so that $f(x) = kx^2$ for $x \leq 3$, and $f(x) = -4x - 6$ for $x > 3$, is continuous at $x = 3$.
- a. -1
 - b. -2
 - c. -3
 - d. 2

- 41 Find k so that $f(x) = kx^2$ for $x \leq 3$, and $f(x) = -4x$ for $x > 3$, is continuous at $x = 3$.
- a. $-\frac{7}{3}$
 - b. $-\frac{1}{3}$
 - c. $\frac{4}{3}$
 - d. $-\frac{4}{3}$
- 42 Find k so that $f(x) = kx^2$ for $x \leq 3$, and $f(x) = -x$ for $x > 3$, is continuous at $x = 3$.
- a. $-\frac{4}{3}$
 - b. $\frac{1}{3}$
 - c. $\frac{2}{3}$
 - d. $-\frac{1}{3}$
- 43 Find k so that $f(x) = kx^2$ for $x \leq -4$, and $f(x) = -4x - 8$ for $x > -4$, is continuous at $x = -4$.
- a. $\frac{1}{2}$
 - b. $-\frac{1}{2}$
 - c. $\frac{3}{2}$
 - d. -2
- 44 Find k so that $f(x) = kx^2$ for $x \leq -1$, and $f(x) = 4x + 6$ for $x > -1$, is continuous at $x = -1$.
- a. -2
 - b. 1
 - c. 2
 - d. 3

45 Find k so that $f(x) = kx^2$ for $x \leq 2$, and $f(x) = -x$ for $x > 2$, is continuous at $x = 2$.

a. -1

b. $-\frac{3}{2}$

c. $-\frac{1}{2}$

d. $\frac{1}{2}$

46 Find k so that $f(x) = kx^2$ for $x \leq 3$, and $f(x) = x - 3$ for $x > 3$, is continuous at $x = 3$.

a. 2

b. -1

c. 0

d. 1

Multiple choice — hard 15

47 Find a so that $f(x) = 0$ for $x \leq 0$, $f(x) = ax + b$ for $0 < x < 3$, and $f(x) = -2$ for $x \geq 3$, is continuous everywhere.

a. $-\frac{2}{3}$

b. $\frac{1}{3}$

c. $-\frac{5}{3}$

d. $\frac{2}{3}$

48 Find a so that $f(x) = 1$ for $x \leq -5$, $f(x) = ax + b$ for $-5 < x < 3$, and $f(x) = 4$ for $x \geq 3$, is continuous everywhere.

a. $-\frac{3}{8}$

b. $\frac{3}{8}$

c. $-\frac{5}{8}$

d. $\frac{11}{8}$

49 Find a so that $f(x) = 3$ for $x \leq -3$, $f(x) = ax + b$ for $-3 < x < 5$, and $f(x) = 4$ for $x \geq 5$, is continuous everywhere.

a. $\frac{9}{8}$

b. $-\frac{7}{8}$

c. $-\frac{1}{8}$

d. $\frac{1}{8}$

50 Find a so that $f(x) = -5$ for $x \leq -3$, $f(x) = ax + b$ for $-3 < x < 5$, and $f(x) = -4$ for $x \geq 5$, is continuous everywhere.

a. $-\frac{7}{8}$

b. $-\frac{1}{8}$

c. $\frac{9}{8}$

d. $\frac{1}{8}$

51 Find a so that $f(x) = 3$ for $x \leq -1$, $f(x) = ax + b$ for $-1 < x < 3$, and $f(x) = 5$ for $x \geq 3$, is continuous everywhere.

a. $\frac{3}{2}$

b. $\frac{5}{2}$

c. $-\frac{1}{2}$

d. $\frac{1}{2}$

52 Find a so that $f(x) = -6$ for $x \leq -1$, $f(x) = ax + b$ for $-1 < x < 1$, and $f(x) = -3$ for $x \geq 1$, is continuous everywhere.

a. $\frac{1}{2}$

b. $\frac{5}{2}$

c. $-\frac{3}{2}$

d. $\frac{3}{2}$

- 53 Find a so that $f(x) = -6$ for $x \leq -2$, $f(x) = ax + b$ for $-2 < x < 1$, and $f(x) = 6$ for $x \geq 1$, is continuous everywhere.
- a. 4
 - b. 5
 - c. 3
 - d. -4
- 54 Find a so that $f(x) = -3$ for $x \leq -4$, $f(x) = ax + b$ for $-4 < x < 5$, and $f(x) = 3$ for $x \geq 5$, is continuous everywhere.
- a. $\frac{2}{3}$
 - b. $-\frac{2}{3}$
 - c. $-\frac{1}{3}$
 - d. $\frac{5}{3}$
- 55 Find a so that $f(x) = 1$ for $x \leq -4$, $f(x) = ax + b$ for $-4 < x < 1$, and $f(x) = 4$ for $x \geq 1$, is continuous everywhere.
- a. $\frac{3}{5}$
 - b. $-\frac{2}{5}$
 - c. $\frac{8}{5}$
 - d. $-\frac{3}{5}$
- 56 Find a so that $f(x) = 2$ for $x \leq 0$, $f(x) = ax + b$ for $0 < x < 1$, and $f(x) = 4$ for $x \geq 1$, is continuous everywhere.
- a. 2
 - b. 1
 - c. -2
 - d. 3

- 57 For $f(x) = x^3 - 4x + 1$, evaluate $f(2)$ and $f(3)$, and state whether the IVT guarantees a root in $(2, 3)$.
- a. $f(2) = 1, f(3) = 16$ — same sign, but a root is still guaranteed since f is a polynomial.
 - b. $f(2) = 1, f(3) = 16$ — same sign, so the IVT does not guarantee a root in $(2, 3)$.
 - c. $f(2) = 1, f(3) = 16$ — the IVT cannot be applied since f is not continuous.
 - d. $f(2) = 1, f(3) = 16$ — opposite signs, so the IVT guarantees a root in $(2, 3)$.
- 58 For $f(x) = x^3 - 4x + 5$, evaluate $f(-3)$ and $f(-2)$, and state whether the IVT guarantees a root in $(-3, -2)$.
- a. $f(-3) = -10, f(-2) = 5$ — the IVT only applies to linear functions, not cubics.
 - b. $f(-3) = -10, f(-2) = 5$ — same sign, so the IVT guarantees no root in $(-3, -2)$.
 - c. $f(-3) = -10, f(-2) = 5$ — opposite signs, so the IVT guarantees a root in $(-3, -2)$.
 - d. $f(-3) = -10, f(-2) = 5$ — opposite signs, but the IVT cannot be applied since f is not continuous.
- 59 For $f(x) = x^3 - 3x + 6$, evaluate $f(-1)$ and $f(0)$, and state whether the IVT guarantees a root in $(-1, 0)$.
- a. $f(-1) = 8, f(0) = 6$ — opposite signs, so the IVT guarantees a root in $(-1, 0)$.
 - b. $f(-1) = 8, f(0) = 6$ — same sign, but a root is still guaranteed since f is a polynomial.
 - c. $f(-1) = 8, f(0) = 6$ — same sign, so the IVT does not guarantee a root in $(-1, 0)$.
 - d. $f(-1) = 8, f(0) = 6$ — the IVT cannot be applied since f is not continuous.
- 60 For $f(x) = x^3 - 6x - 1$, evaluate $f(0)$ and $f(1)$, and state whether the IVT guarantees a root in $(0, 1)$.
- a. $f(0) = -1, f(1) = -6$ — opposite signs, so the IVT guarantees a root in $(0, 1)$.
 - b. $f(0) = -1, f(1) = -6$ — same sign, but a root is still guaranteed since f is a polynomial.
 - c. $f(0) = -1, f(1) = -6$ — the IVT cannot be applied since f is not continuous.
 - d. $f(0) = -1, f(1) = -6$ — same sign, so the IVT does not guarantee a root in $(0, 1)$.

- 61** For $f(x) = x^3 + 6x + 1$, evaluate $f(-1)$ and $f(0)$, and state whether the IVT guarantees a root in $(-1, 0)$.
- a.** $f(-1) = -6$, $f(0) = 1$ — opposite signs, but the IVT cannot be applied since f is not continuous.
 - b.** $f(-1) = -6$, $f(0) = 1$ — same sign, so the IVT guarantees no root in $(-1, 0)$.
 - c.** $f(-1) = -6$, $f(0) = 1$ — the IVT only applies to linear functions, not cubics.
 - d.** $f(-1) = -6$, $f(0) = 1$ — opposite signs, so the IVT guarantees a root in $(-1, 0)$.